

Лабораторная работа №6. Основы интерфейса взаимодействия пользователя с системой Unix на уровне командной строки

Чжан Вэньцзюнь

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1. Цель работы

Приобретение практических навыков работы с Unix через командную строку.

2. Ход работы

2.1 Определение текущего каталога

```
(base) hhl@LAPTOP-058L8DAA:~$ pwd  
/home/hhl
```

Команда: pwd

Результат: /home/wenjun/work/study/2026-2027/Операционные системы/os-intro/labs/lab06

2.2 Работа с каталогом /tmp

Переход в /tmp

```
(base) hhl@LAPTOP-058L8DAA:~$ cd /tmp
```

Команды: cd /tmp pwd

Просмотр содержимого

```
(base) hhl@LAPTOP-058L8DAA:/tmp$ ls
snap-private-tmp
systemd-private-07c39e09d0c340809ball14be17f1acc0-polkit.service-k0DwV5
systemd-private-07c39e09d0c340809ball14be17f1acc0-system-logind.service-2VshhC
systemd-private-07c39e09d0c340809ball14be17f1acc0-system-resolved.service-CpmJMS
systemd-private-07c39e09d0c340809ball14be17f1acc0-system-timesyncd.service-sy2B0Z
systemd-private-07c39e09d0c340809ball14be17f1acc0-wsl-pro.service-1yY6Lo
(base) hhl@LAPTOP-058L8DAA:/tmp$ ls -l
total 24
drwxr-xr-x 2 root root 4096 Mar 19 00:05 snap-private-tmp
drwxr-xr-x 3 root root 4096 Mar 19 00:06 systemd-private-07c39e09d0c340809ball14be17f1acc0-polkit.service-k0DwV5
drwxr-xr-x 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340809ball14be17f1acc0-system-logind.service-2VshhC
drwxr-xr-x 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340809ball14be17f1acc0-system-resolved.service-CpmJMS
drwxr-xr-x 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340809ball14be17f1acc0-system-timesyncd.service-sy2B0Z
drwxr-xr-x 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340809ball14be17f1acc0-wsl-pro.service-1yY6Lo
(base) hhl@LAPTOP-058L8DAA:/tmp$ ls -a
.
..
snap-private-tmp
systemd-private-07c39e09d0c340809ball14be17f1acc0-polkit.service-k0DwV5
systemd-private-07c39e09d0c340809ball14be17f1acc0-system-logind.service-2VshhC
systemd-private-07c39e09d0c340809ball14be17f1acc0-system-resolved.service-CpmJMS
systemd-private-07c39e09d0c340809ball14be17f1acc0-system-timesyncd.service-sy2B0Z
systemd-private-07c39e09d0c340809ball14be17f1acc0-wsl-pro.service-1yY6Lo
(base) hhl@LAPTOP-058L8DAA:/tmp$ ls -la
total 40
drwxr-xr-x 9 root root 12288 Mar 19 00:06 .
drwxr-xr-x 22 root root 4096 Mar 19 00:05 ..
drwxr-xr-x 2 root root 60 Mar 19 00:05 .
drwxr-xr-x 2 root root 4096 Mar 19 00:05 snap-private-tmp
drwxr-xr-x 3 root root 4096 Mar 19 00:06 systemd-private-07c39e09d0c340809ball14be17f1acc0-polkit.service-k0DwV5
drwxr-xr-x 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340809ball14be17f1acc0-system-logind.service-2VshhC
drwxr-xr-x 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340809ball14be17f1acc0-system-resolved.service-CpmJMS
drwxr-xr-x 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340809ball14be17f1acc0-system-timesyncd.service-sy2B0Z
drwxr-xr-x 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340809ball14be17f1acc0-wsl-pro.service-1yY6Lo
(base) hhl@LAPTOP-058L8DAA:/tmp$ ls -F
snap-private-tmp/
systemd-private-07c39e09d0c340809ball14be17f1acc0-polkit.service-k0DwV5/
systemd-private-07c39e09d0c340809ball14be17f1acc0-system-logind.service-2VshhC/
systemd-private-07c39e09d0c340809ball14be17f1acc0-system-resolved.service-CpmJMS/
systemd-private-07c39e09d0c340809ball14be17f1acc0-system-timesyncd.service-sy2B0Z/
systemd-private-07c39e09d0c340809ball14be17f1acc0-wsl-pro.service-1yY6Lo/
```

Команды: ls ls -l ls -a ls -la ls -F

Описание: ls — список файлов ls -l — подробный вывод ls -a — скрытые файлы ls -la — полный список ls -F — типы файлов

Проверка системных каталогов

```
(base) hhl@LAPTOP-058L8DAA:/tmp$ ls /var/spool
cron libreoffice mail rsyslog
```

Команда: ls /var/spool

Возврат в рабочий каталог

```
(base) hhl@LAPTOP-058L8DAA:/tmp$ cd ~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ pwd
/home/hhl/work/study/2026-2027/Операционные системы/os-intro/labs/lab06
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ ls -l
total 8
drwxr-xr-x 4 hhl hhl 4096 Feb 22 04:03 presentation
drwxr-xr-x 6 hhl hhl 4096 Feb 22 04:03 report
```

Команды: cd ~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06 pwd ls -l

2.3 Работа с каталогами

Создание каталога

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ mkdir newdir
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ ls -ld newdir
drwxr-xr-x 2 hhl hhl 4096 Mar 19 00:15 newdir
```

Команды: mkdir newdir ls -ld newdir

Подкаталог

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ mkdir newdir/morefun
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ ls newdir
morefun
```

Команды: `mkdir newdir/morefun` `ls newdir`

Несколько каталогов

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ mkdir letters memos misk
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ ls
letters memos misk newdir presentation report
```

Команды: `mkdir letters memos misk` `ls`

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ rmdir letters memos misk
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ ls
newdir presentation report
```

Команды: `rmdir letters memos misk` `ls`

Ошибка удаления каталога

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ rm newdir
rm: cannot remove 'newdir': Is a directory
```

Команда: `rm newdir`

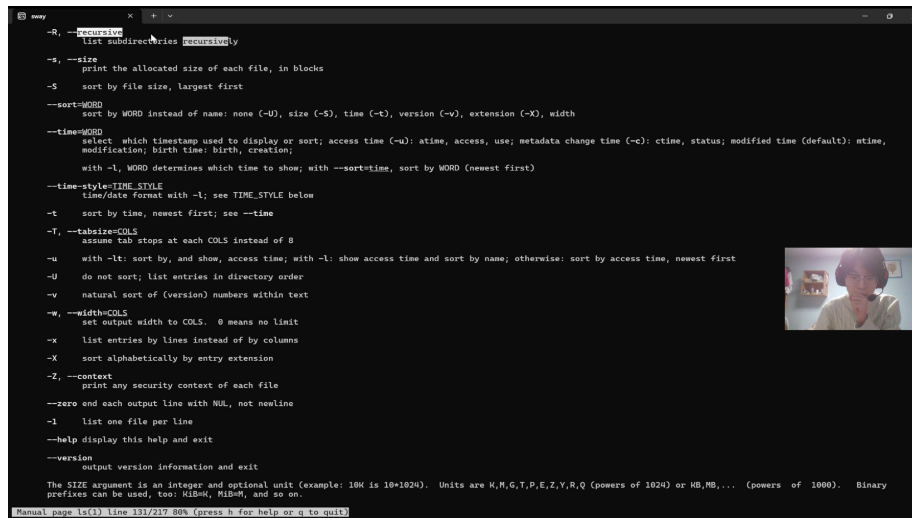
Ошибка: `rm: cannot remove 'newdir': Is a directory`

Удаление подкаталога

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ rmdir newdir/morefun
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ ls newfir
ls: cannot access 'newfir': No such file or directory
```

Команды: `rmdir newdir/morefun` `ls newdir`

2.4 Рекурсивный вывод



```
-R, --recursive
    list subdirectories recursively

-s, --size
    print the allocated size of each file, in blocks

-S
    sort by file size, largest first

--sort=WORD
    sort by WORD instead of name (-U), size (-S), time (-t), version (-v), extension (-X), width

--time=WORD
    select which timestamp used to display or sort; access time (-u): atime, access, use; metadata change time (-c): ctime, status; modified time (default): mtime,
    modification; birth time: birth, creation;
    with -l, WORD determines which time to show; with --sort=time, sort by WORD (newest first)

--time-style=TIME_STYLE
    time/date format with -l; see TIME_STYLE below

-t
    sort by time, newest first; see --time

-T, --tabsize=COLS
    assume tab stops at each COLS instead of 8

-u
    with -lt: sort by, and show, access time; with -l: show access time and sort by name; otherwise: sort by access time, newest first

-U
    do not sort; list entries in directory order

-v
    natural sort of (version) numbers within text

-w, --width=COLS
    set output width to COLS. 0 means no limit

-x
    list entries by lines instead of by columns

-X
    sort alphabetically by entry extension

-Z, --context
    print any security context of each file

--zero
    end each output line with NUL, not newline

-l
    list one file per line

--help
    display this help and exit

--version
    output version information and exit

The SIZE argument is an integer and optional unit (example: 10K is 10*1024). Units are K,M,G,T,P,E,Z,Y,R,Q (powers of 1024) or KB,MB,... (powers of 1000). Binary
    prefixes can be used, too: KiB,MiB,... and so on.

Manual page ls(1) line 131/217 88% (press h for help or q to quit)
```

Команда: `ls -R`

2.5 Сортировка

Команды: `ls -lt` `ls -ltr`

2.6 Основные команды

`cd`: `cd ~` `cd ..` `cd -`

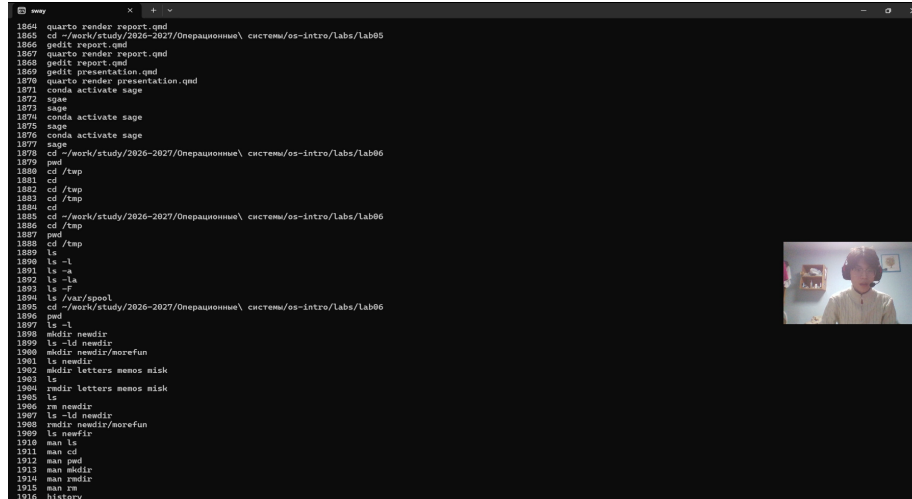
`pwd`: `pwd -L` `pwd -P`

`mkdir`: `mkdir -p dir1/dir2/dir3` `mkdir -m 755 dir`

`rmdir`: `rmdir -p dir1/dir2/dir3`

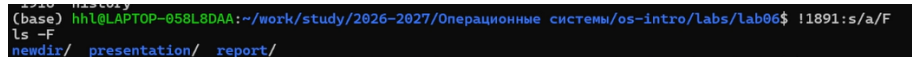
`rm`: `rm file.txt` `rm -r dir` `rm -rf dir`

2.7 История команд



```
1864 quarto render report.qmd
1865 cd ~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06
1866 gedit report.qmd
1867 quarto render report.qmd
1868 gedit presentation.qmd
1869 quarto render presentation.qmd
1870 conda activate sage
1871 sage
1872 conda activate sage
1873 sage
1874 cd ~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06
1875 pwd
1876 cd /tmp
1877 cd
1878 cd /tmp
1879 cd /tmp
1880 cd
1881 cd ~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06
1882 cd /tmp
1883 cd /tmp
1884 cd
1885 cd /tmp
1886 ls
1887 ls -l
1888 ls -a
1889 ls -la
1890 ls -F
1891 ls /var/spool
1892 cd ~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06
1893 pwd
1894 ls -l
1895 mkdir newdir
1896 ls -ld newdir
1897 mkdir newdir/morefun
1898 ls newdir
1899 mkdir letters memos misk
1900 ls
1901 rm newdir
1902 ls -ld newdir
1903 rm newdir/morefun
1904 ls newdir
1905 man ls
1906 man cd
1907 man pwd
1908 man mkdir
1909 man rm
1910 history
```

Команда: history



```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ !1891:s/a/F
ls -F
newdir/ presentation/ report/
```

Команды: !1891 !1891:s/a/F

3. Контрольные вопросы

Командная строка — текстовый интерфейс управления системой.

Команды: pwd ls -F ls -a ls -la rm file rmdir dir rm -r dir history

Дополнительно: cd /absolute/path cd relative/path mkdir dir && cd dir cd bad || echo "error" touch My file.txt echo "Price \$100"

4. Вывод

Освоены базовые команды Unix, работа с каталогами, навигация по файловой системе и история команд.